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- Gypsum is used in toothpastes, shampoos, and hair products, mainly due to its binding and thickening properties.

- Gypsum is a component of Portland cement, where it acts as a hardening retarder to control the speed at which concrete sets.

- To sum up, gypsum is one of the most abundant minerals that have endless uses and applications. Mining of gypsum is simple and easy, as the mineral occurs in large thick beds near the Earth's surface. However, large-scale mining of gypsum involves considerable damage to the environment. Gypsum can also be recycled, but not much importance has been given to recycle this mineral due to its abundance.



Figure 5.13 -**Uses of Gypsum**

5.6.4 Plaster of paris

Calcium Sulphate (Plaster of Paris), $CaSO_4 \cdot ½\ H_2O$

It is a hemihydrate of calcium sulphate. It is obtained when gypsum, $CaSO_4 \cdot 2H_2O$, is heated to 393 K.

$$2CaSO_4 .2H_2O(s) \rightarrow 2CaSO_4 .H_2O + 3H_2O$$

Above 393 K, no water of crystallisation is left and anhydrous calcium sulphate, $CaSO_4$ is formed. This is known as 'dead burnt plaster'.